



WORKPLACE NINJAS UK

Endpoint Privilege Management and App Control for Business: Friends or Foes?

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A brief recap of EPM

The elephant in the room...

Intune Plan 1

Included in EMS E3 or Microsoft 365 E3, ME5, F1, F3, and Business Premium plans

- Cross-platform endpoint management
- Endpoint security built in
- Endpoint analytics
- App protection policies
- Shared device management
- Device compliance
- Risk and app based conditional access
- Proactive remediation



Intune Suite

Add to Plan 1 to utilize these solutions*

- Remote Help
- **Endpoint Privilege Management**
- Advanced Analytics
- Enterprise Application Management
- Cloud PKI
- [All Intune Plan 2 features](#)

Prerequisite

- Intune Plan 1

Intune Plan 2

Add to Plan 1 to utilize these features

- Tunnel for Mobile App Management
- Mobile firmware update management
- Specialty device management
- Multiple managed accounts (future)

Prerequisite

- Intune Plan 1



**Also available as individual add-ons
aka.ms/IntuneSuitePricing*

What is the problem?

Running as a standard user reduces the attack surface...

- Zero Trust Framework
 - Least Privilege Access

Impact on user productivity...

- Standard users can't do everything



How does it work?

EPM Elevated Account

The moment EPM gets installed, a new virtual admin account

Creating the Virtual Account

To create the virtual account it uses the LsaManageSidNameMapping

A Virtual Domain?

Besides the virtual account, a virtual domain: MEM will be created (requirement to create the account)

Virtual Account Created

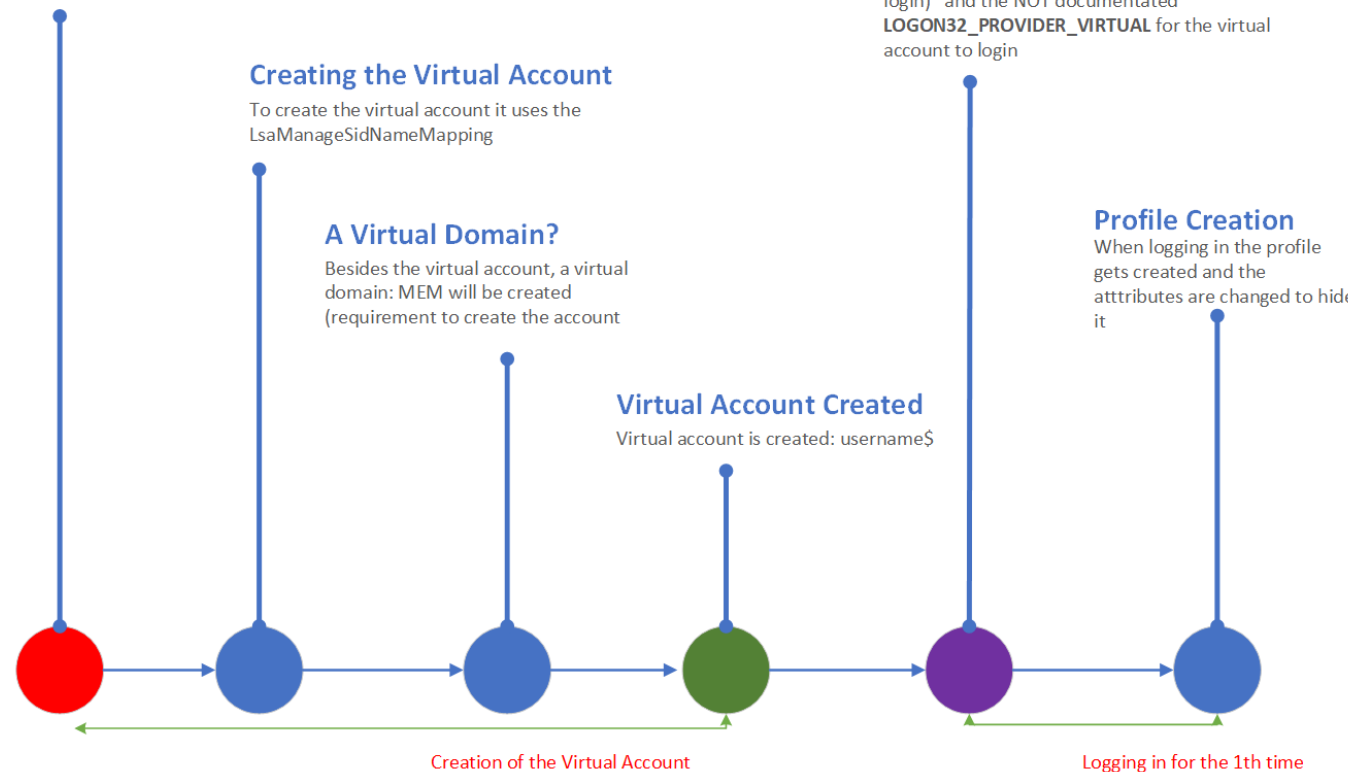
Virtual account is created: username\$

Virtual Account Login

The virtual account uses the LogonUserExEx API (NEEDS DOMAIN (MEM) AND USERNAME to login) and the NOT documented LOGON32_PROVIDER_VIRTUAL for the virtual account to login

Profile Creation

When logging in the profile gets created and the attributes are changed to hide it



Source: Ooms, R. (2023, May 31). The Virtual Account that rocks the EPM! Call4Cloud from <https://call4cloud.nl/virtual-account-epm-elevation/>

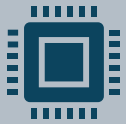




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A short introduction of App Control for Business

What is App Control for Business?



Windows Defender Application Control (WDAC) is now known as App Control for Business



Face the challenges of malicious files



Designed to manage and prevent undesired apps



Overview of App Control for Business



Uses the Windows **Application Control CSP** to manage which apps are allowed to run on Windows devices.



Leverages the Managed Installer to ensure only trusted applications can run on your Windows devices



How does it work?

IT admin **creates** Managed Installer policy to add **Intune Management Extension** as **Managed Installer** for tenant wide

IT admin **deploys** an **app** via Intune

IT admin deploys an **App Control for Business** policy via Intune

Service Side

Tagged apps can be identified by your App Control for Business policies as safe apps that can be **allowed to run** on your devices.

The **apps** you deploy through Intune are automatically **tagged** by the **Managed installer**

Client Side



Workflow

IT admin deploys
Managed Installer policy

IT admin deploys an App
Control for Business
policy via Intune



1

2

3



Policy starts getting applied to
Windows endpoints

Apps will start getting tagged

IT admin deploys
app via Intune

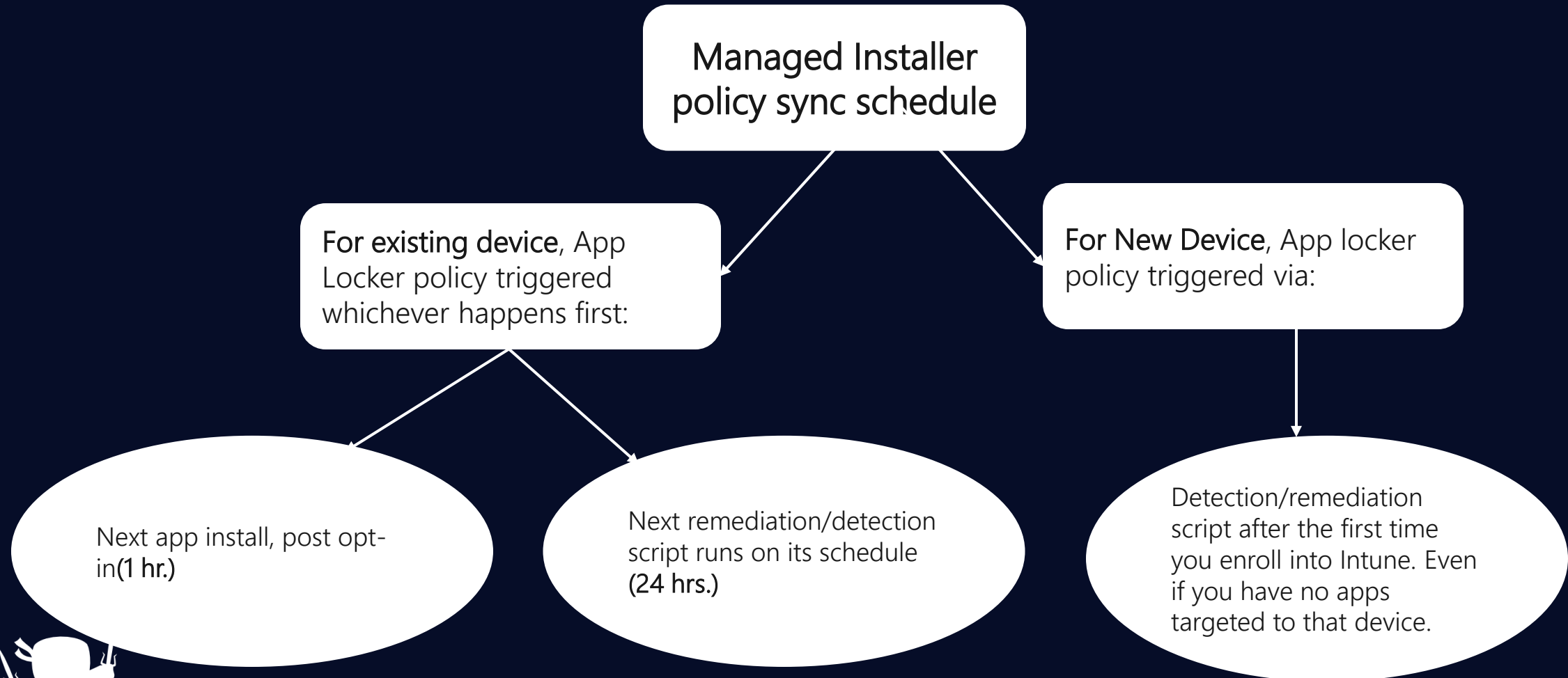
4



End users can only launch apps that
are part of the app allow list.
If **“trust apps from managed
installer”** rule is present, it will
allow tagged apps to run



Policy Sync Schedule



Creating App Control for Business policies

Home > Endpoint security

Endpoint security | App Control for Business (Preview)

Search

App Control for Business Managed installer

Overview

- Overview
- All devices
- Security baselines
- Security tasks

Manage

- Antivirus
- Disk encryption
- Firewall
- Endpoint Privilege Management
- Endpoint detection and response
- App Control for Business (Preview)**
- Attack surface reduction
- Account protection
- Device compliance
- Conditional access

Intune Management Extension is set as an authorized source of application deployment (managed installer) for your organization. [Learn more about managed installer and App Control for Business.](#)

+ Create Policy Refresh Export

Search by profile name

Policy name	↑↓	Policy type	↑↓	Assigned	↑↓	Platform	↑↓	Target	↑↓	Last modified	↑↓
App Control Policy		App Control for Business		No		Windows 10 and later		mdm		07/04/24, 1:29 PM	...



Getting started with App Control for Business policies

Using built-in controls

- Enable trust of Windows components and store apps
- Trust apps with a good reputation
- Trust apps from **managed installers**

Enter xml data

- For custom properties definition
- Use built-in controls for easy configuration
 - App Control Wizard!



Getting started with App Control for Business policies

[Home](#) > [Endpoint security | App Control for Business \(Preview\)](#) > [App Control Policy](#) >

Edit profile - App Control Policy ...

Settings catalog

1 Configuration settings **2** Review + save

^ App Control for Business

Configuration settings format  

**App Control for Business
Built In Controls**

Enable App Control for Business policy to trust Windows components and Store apps  * 

Select additional rules for trusting apps  

Use built-in controls 

Enter xml data

Use built-in controls

Enforce 

2 selected 



Getting started with App Control for Business policies

[Home](#) > [Endpoint security | App Control for Business \(Preview\)](#) >

Create profile ...

App Control for Business

✓ Basics **2 Configuration settings** ③ Scope tags ④ Assignments ⑤ Review + create

^ App Control for Business

Configuration settings format ⓘ Use built-in controls ▾

App Control for Business Built In Controls

Enable App Control for Business policy to trust Windows components and Store apps * ⓘ Audit only ▾

Select additional rules for trusting apps ⓘ 2 selected ▾

- ☒ Trust apps with good reputation
- ☒ Trust apps from managed installers



Getting started with App Control for Business policies

[Home](#) > [Endpoint security](#) | [App Control for Business \(Preview\)](#) >

Create profile ...

App Control for Business

✓ Basics **2 Configuration settings** ③ Scope tags ④ Assignments ⑤ Review + create

^ App Control for Business

Configuration settings format ⓘ

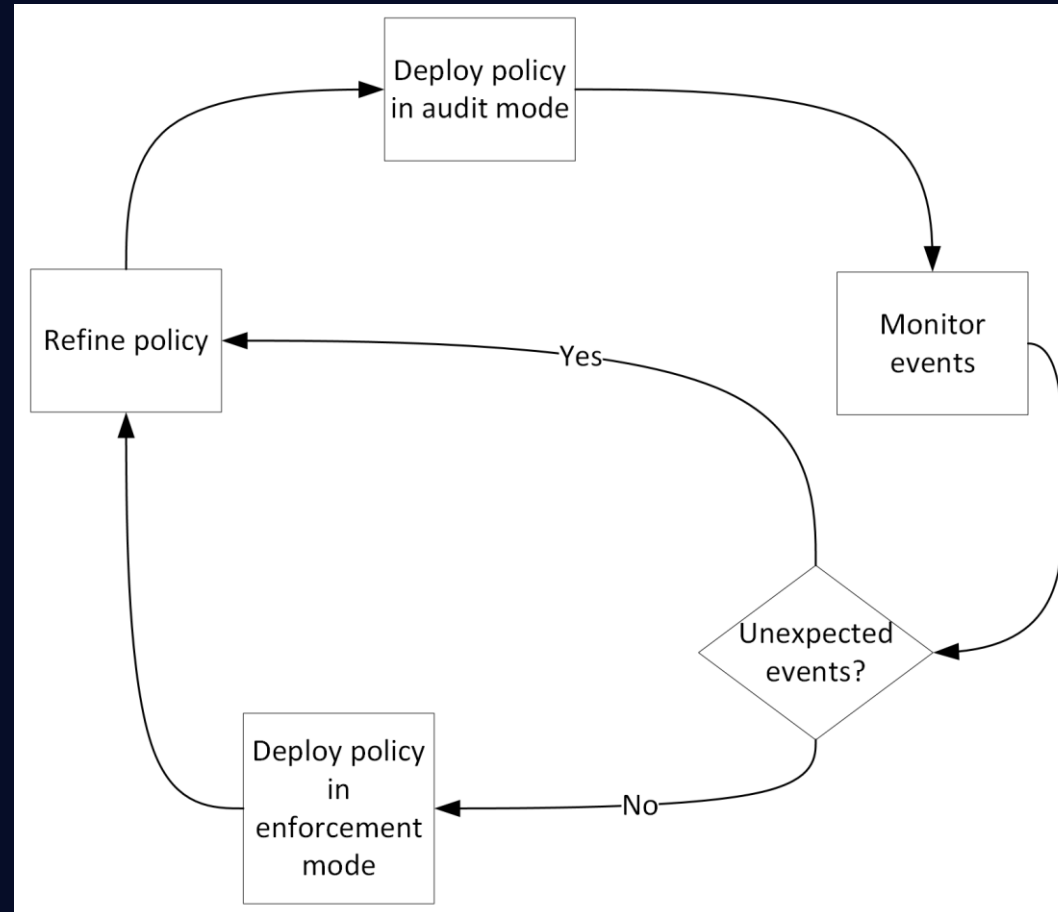
App Control for Business policy ⓘ 

XML value *

Not configured



Creating the XML



App Control Policy Wizard



[Home](#) [About](#) [Archives](#) [Contact](#)

App Control Policy Wizard

The Microsoft App Control Wizard (Version 2.6.0.1) enables IT and security professionals to build and edit Application Control for Business (formerly WDAC) policies by providing an easy to use interface.

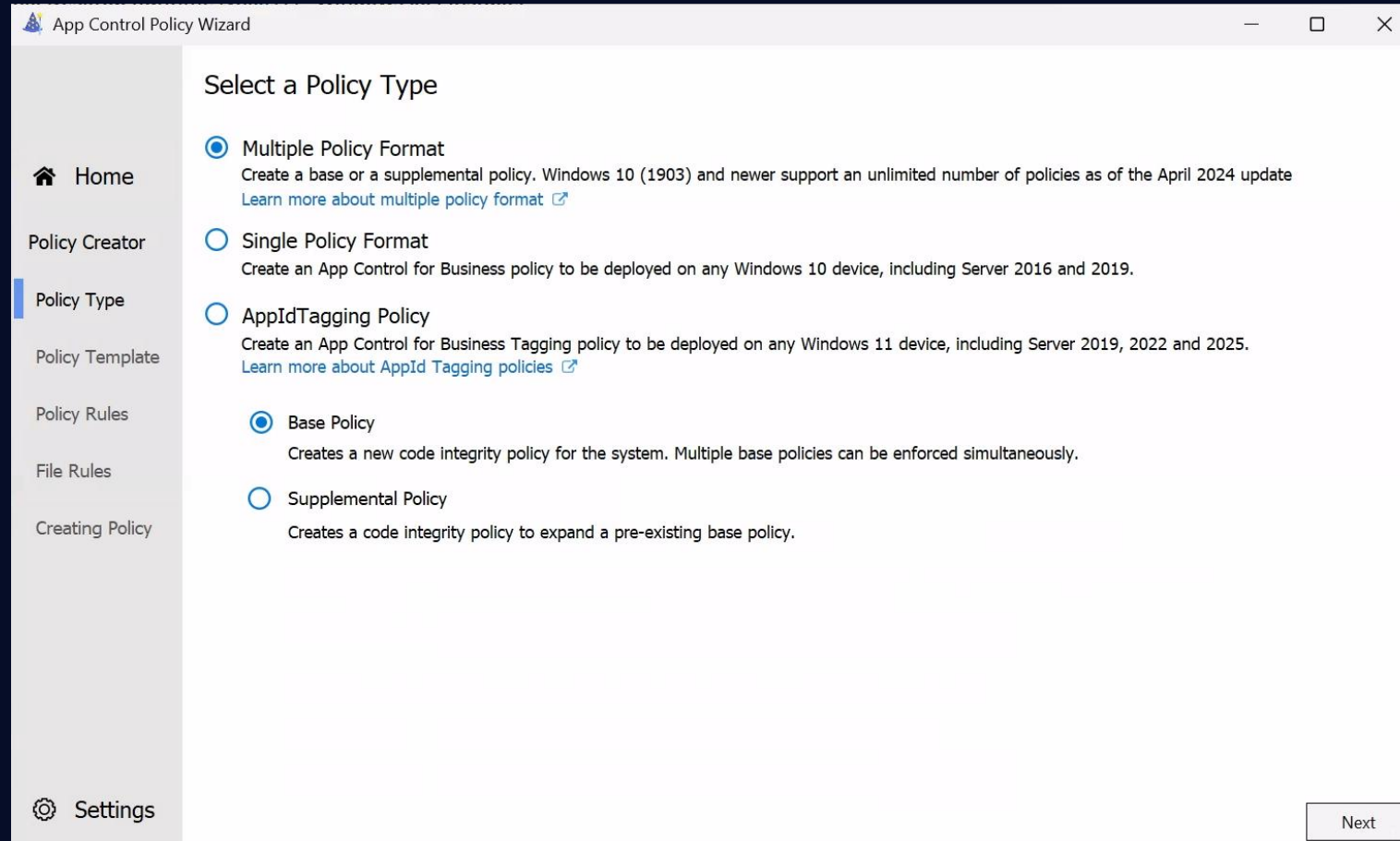
Use this application to create, edit and merge new base and supplemental App Control policies, in addition to converting event logs data to policies.

Getting Started

You can install the Policy Wizard by selecting the download link. Before you install the application:



Creating a Base Policy



The screenshot shows the 'App Control Policy Wizard' window. On the left is a sidebar with navigation links: Home, Policy Creator, Policy Type (highlighted), Policy Template, Policy Rules, File Rules, and Creating Policy. At the bottom of the sidebar is a 'Settings' link. The main area is titled 'Select a Policy Type' and contains three radio button options:

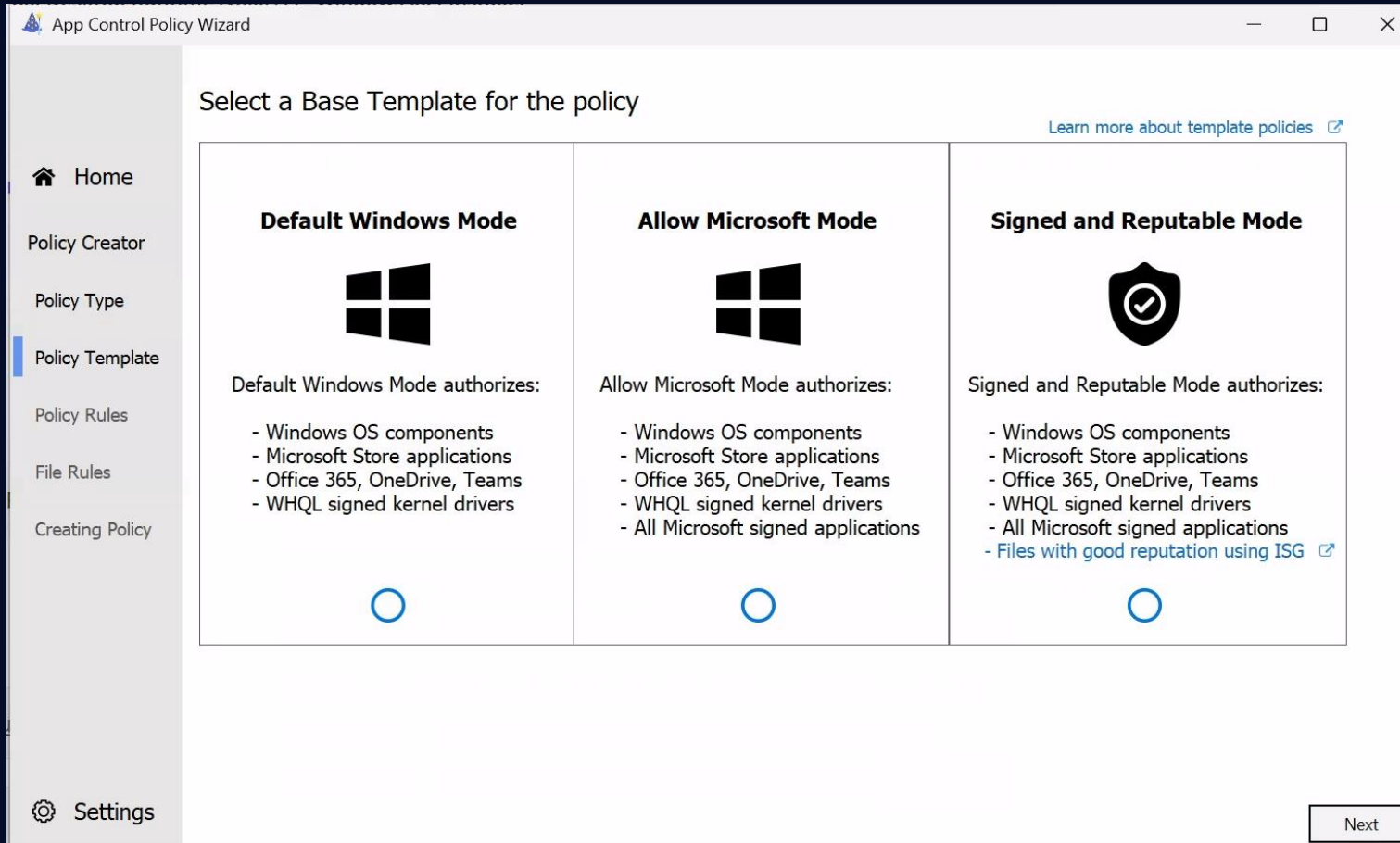
- ☒ **Multiple Policy Format**
Create a base or a supplemental policy. Windows 10 (1903) and newer support an unlimited number of policies as of the April 2024 update.
[Learn more about multiple policy format](#)
- ☐ **Single Policy Format**
Create an App Control for Business policy to be deployed on any Windows 10 device, including Server 2016 and 2019.
- ☐ **AppIdTagging Policy**
Create an App Control for Business Tagging policy to be deployed on any Windows 11 device, including Server 2019, 2022 and 2025.
[Learn more about AppId Tagging policies](#)

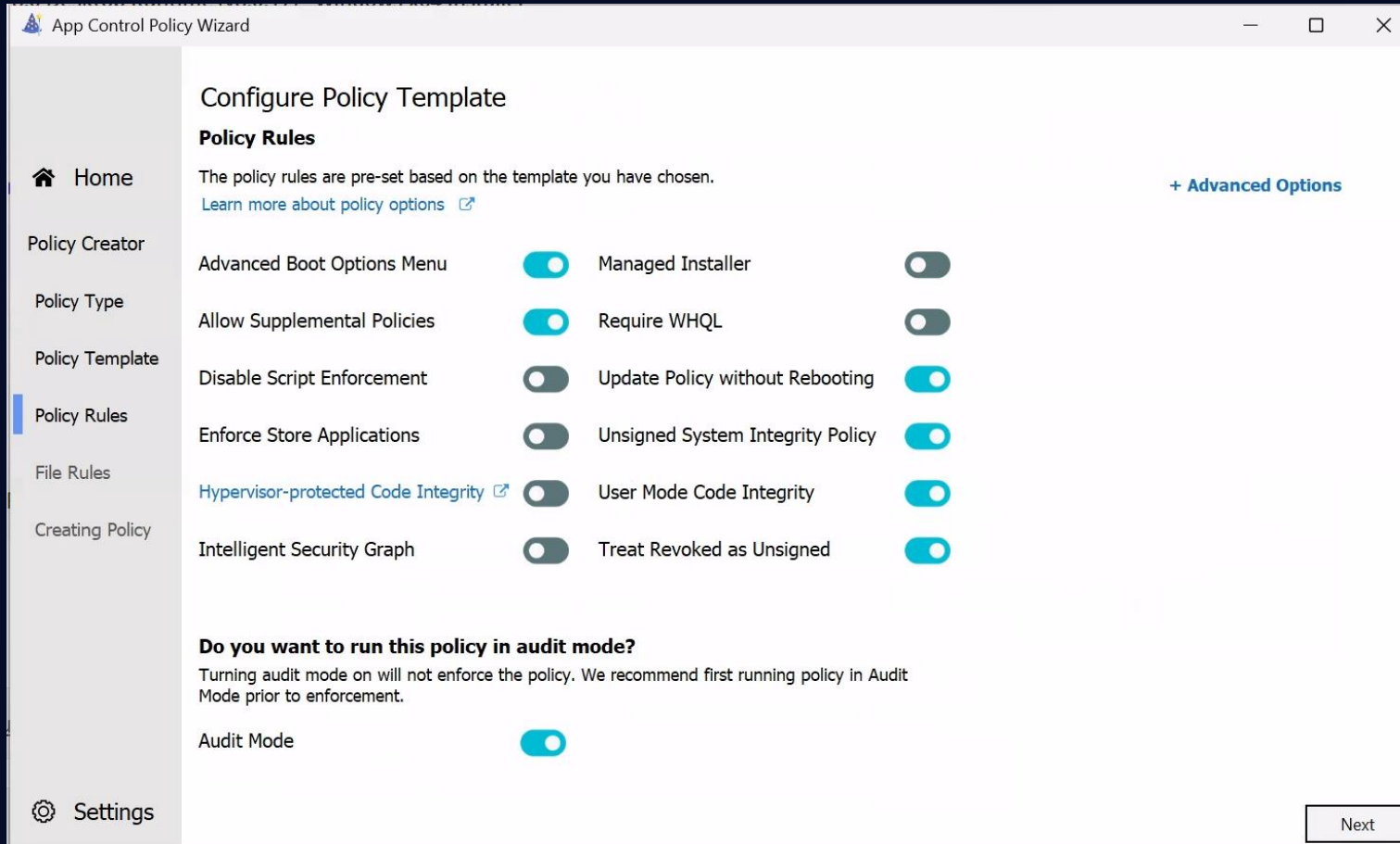
Below these options, there are two more radio button options:

- ☒ **Base Policy**
Creates a new code integrity policy for the system. Multiple base policies can be enforced simultaneously.
- ☐ **Supplemental Policy**
Creates a code integrity policy to expand a pre-existing base policy.

A 'Next' button is located at the bottom right of the window.







Using a supplemental policy



Policy Expansion

Enhances base policy to include more trusted applications

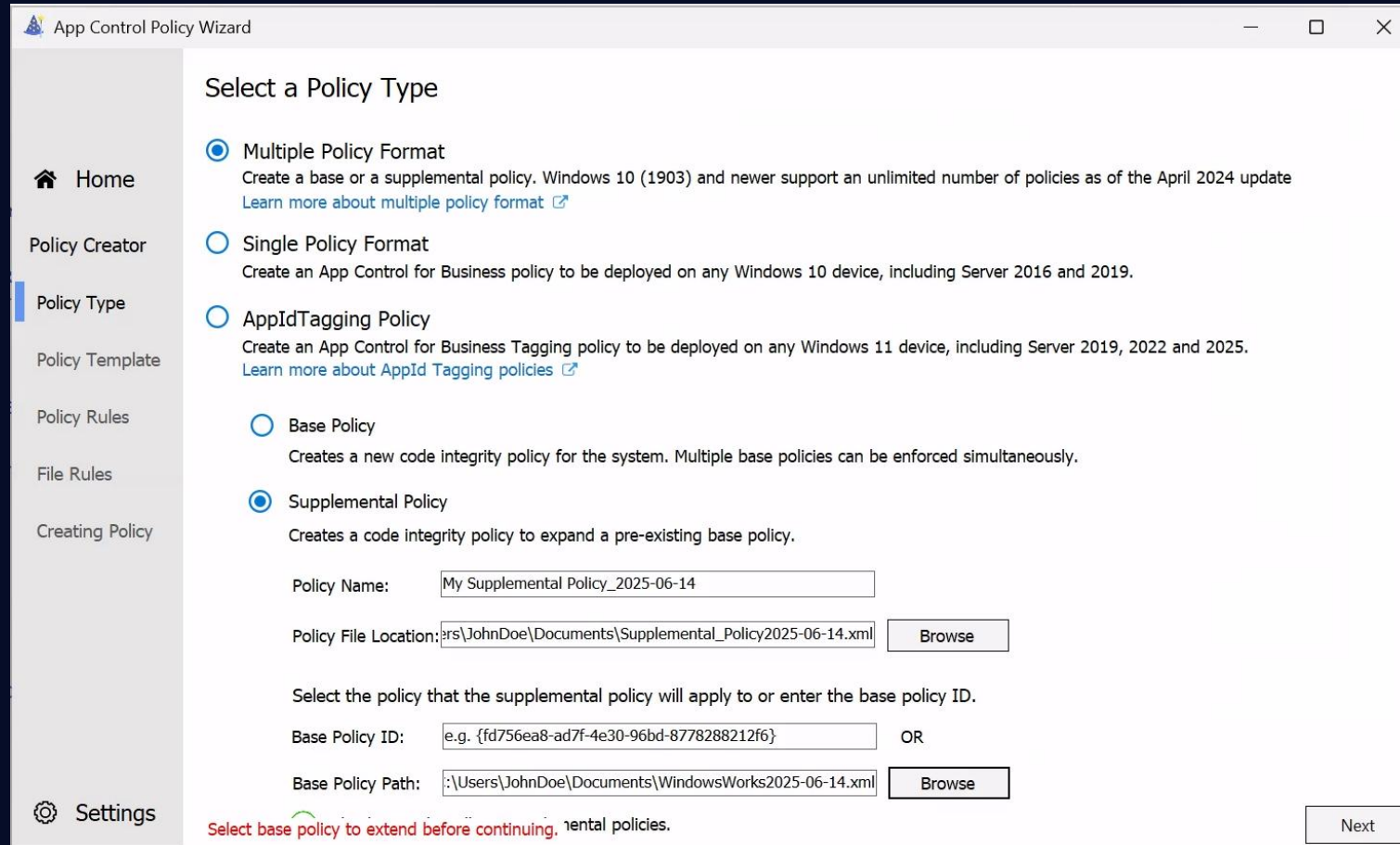


Multiple Supplementals for One Base Policy

Allows various apps for different teams



Creating a Supplemental Policy



The image shows a screenshot of the 'App Control Policy Wizard' window. The window has a title bar with the text 'App Control Policy Wizard' and standard window controls. On the left is a sidebar with navigation links: 'Home', 'Policy Creator', 'Policy Type' (highlighted), 'Policy Template', 'Policy Rules', 'File Rules', 'Creating Policy', and 'Settings'. The main area is titled 'Select a Policy Type' and contains several radio button options. The 'Supplemental Policy' option is selected. Below this, there are input fields for 'Policy Name', 'Policy File Location', 'Base Policy ID', and 'Base Policy Path', each with a 'Browse' button. A 'Next' button is at the bottom right. A red warning message is at the bottom left of the main area.

App Control Policy Wizard

Select a Policy Type

- ☒ Multiple Policy Format
Create a base or a supplemental policy. Windows 10 (1903) and newer support an unlimited number of policies as of the April 2024 update.
[Learn more about multiple policy format](#)
- ☐ Single Policy Format
Create an App Control for Business policy to be deployed on any Windows 10 device, including Server 2016 and 2019.
- ☐ AppIdTagging Policy
Create an App Control for Business Tagging policy to be deployed on any Windows 11 device, including Server 2019, 2022 and 2025.
[Learn more about AppId Tagging policies](#)
- ☐ Base Policy
Creates a new code integrity policy for the system. Multiple base policies can be enforced simultaneously.
- ☒ Supplemental Policy
Creates a code integrity policy to expand a pre-existing base policy.

Policy Name:

Policy File Location:

Select the policy that the supplemental policy will apply to or enter the base policy ID.

Base Policy ID: OR

Base Policy Path:

Select base policy to extend before continuing. ental policies.



Using a Managed Installer

App Control for Business

Managed installer

To begin configuring the Intune Management Extension as a Managed Installer on all applicable enrolled devices in the tenant, select "Add". You can review the status of the policy in your tenant, and the number of devices where the policy succeeded or resulted in an error. [Learn more about how Intune sets the managed installer.](#)

 Add  Refresh  Export  Columns

Add managed installer

App Control for Business

Add will grant Microsoft permission to configure Intune Management Extension as a managed installer (an authorized source for application deployment) on applicable devices.

If necessary, you can edit the policy to pause any scheduled policy that sets the Intune Management Extension on applicable devices. Then, any policy won't be changed. If you need to rerun the cleanup script. [Learn more about how to manage the managed installer.](#)

Next, you can create an App Control policy to define the rules for which apps can run on your devices. [Learn more about managed installers and App Control.](#)

Add

Add managed installer

If you grant permission to Microsoft to configure Intune Management Extension as a managed installer, all apps installed from this source can be configured as trusted in your App Control for Business policy.

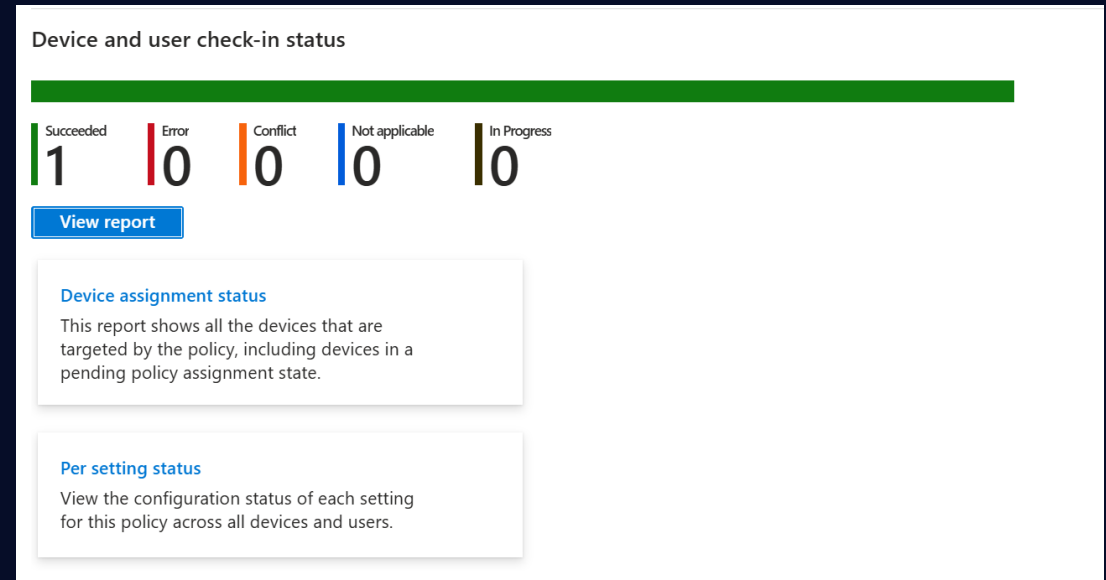
Yes

No



Monitoring App Control for Business policies

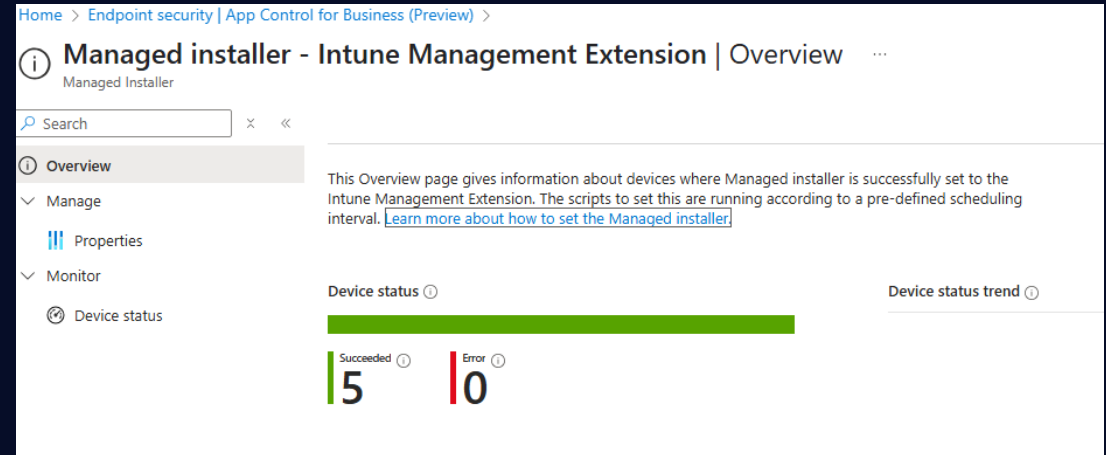
- Verify the status of the policy assignment
- Verify the status of the Managed Installer



Monitoring App Control for Business policies

- Verify the status of the policy assignment
- **Verify the status of the Managed Installer**

What about the audit mode logs locally on the client?



Troubleshooting from the endpoint

- Using Registry
- PowerShell
- File Explorer
- Event Viewer
- Intune Management Extension logs
- KQL (Device Query)



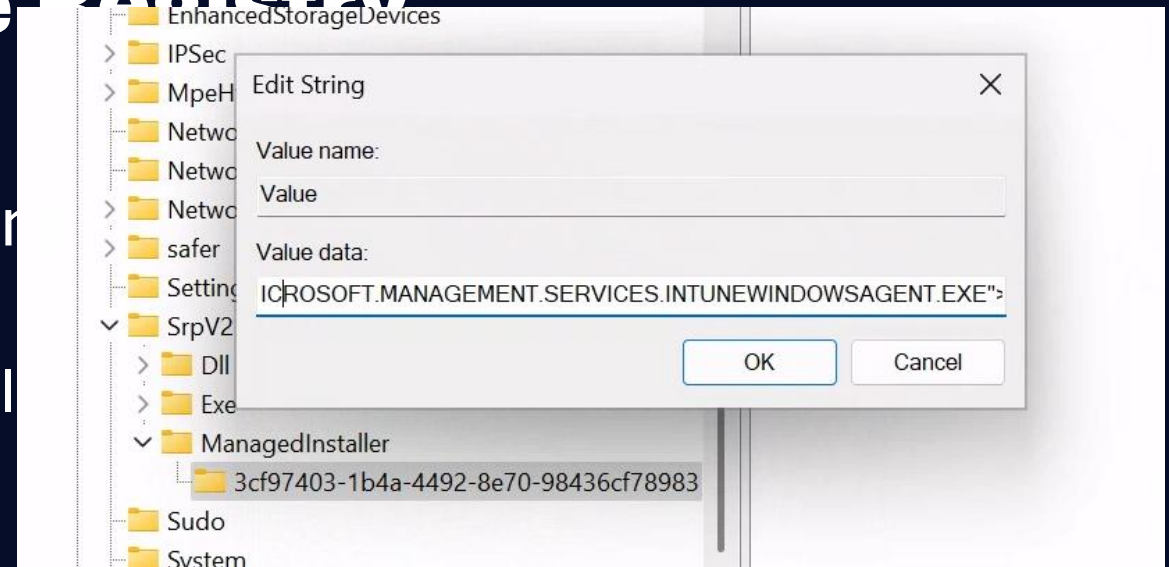
Troubleshoot using the Registry

- Check if the Intune Management
Installer

- Computer\HKEY_LOCAL_MACHINE\IntuneManagementExtension\SideCarPolicies
- ManagedInstallerEnabled = 1

- Verify that the **Managed Installer** policy from Intune delivered

- Computer\HKEY_LOCAL_MACHINE\SOFTWARE\Policies\Microsoft\Windows\SrpV2\ManagedInstaller



Troubleshoot using PowerShell

- Export the details of the AppLocker policy to an XML file
 - Get-AppLockerPolicy - Effective -Xml | Set-Content ("C:\TEMP\WPNinjas.xml")

A screenshot of a web browser displaying the XML content of a file named WPNinjas.xml. The browser's address bar shows the file path C:/TEMP/WPNinjas.xml. The XML content is an AppLocker policy configuration. It includes three rule collections: a 'Dll' rule for blocking a DLL file, an 'Exe' rule for blocking an executable file, and a 'ManagedInstaller' rule for allowing a specific Microsoft management service. The XML is well-formatted with indentation and color-coding for tags and attributes.

```
<?xml version="1.0" encoding="utf-8" standalone="yes" ?>
<AppLockerPolicy Version="1">
  <RuleCollection Type="Dll" EnforcementMode="AuditOnly">
    <FilePathRule Id="86f235ad-3f7b-4121-bc95-ea8bde3a5db5" Name="Dummy Rule" Description="" UserOrGroupSid="S-1-1-0" Action=
      <Conditions>
        <FilePathCondition Path="%OSDRIVE%\ThisWillBeBlocked.dll"/>
      </Conditions>
    </FilePathRule>
    <RuleCollectionExtensions>
      <ThresholdExtensions>
        <Services EnforcementMode="Enabled"/>
      </ThresholdExtensions>
      <RedstoneExtensions/>
    </RuleCollectionExtensions>
  </RuleCollection>
  <RuleCollection Type="Exe" EnforcementMode="AuditOnly">
    <FilePathRule Id="9420c496-046d-45ab-bd0e-455b2649e41e" Name="Dummy Rule" Description="" UserOrGroupSid="S-1-1-0" Action=
      <Conditions>
        <FilePathCondition Path="%OSDRIVE%\ThisWillBeBlocked.exe"/>
      </Conditions>
    </FilePathRule>
    <RuleCollectionExtensions>
      <ThresholdExtensions>
        <Services EnforcementMode="Enabled"/>
      </ThresholdExtensions>
      <RedstoneExtensions/>
    </RuleCollectionExtensions>
  </RuleCollection>
  <RuleCollection Type="ManagedInstaller" EnforcementMode="AuditOnly">
    <FilePublisherRule Id="3cf97403-1b4a-4492-8e70-98436cf78983" Name="MICROSOFT.MANAGEMENT.SERVICES.INTUNEWINDOWSAGENT.EXE v
      Action="Allow">
      <Conditions>
        <FilePublisherCondition PublisherName="O=MICROSOFT CORPORATION, L=REDMOND, S=WASHINGTON, C=US" ProductName="" Binary
          <BinaryVersionRange LowSection="1.37.200.8" HighSection=""/>
        </FilePublisherCondition>
      </Conditions>
    </FilePublisherRule>
  </RuleCollection>
</AppLockerPolicy>
```


Troubleshoot using PowerShell

- Verify the **Managed Installer** tag
 - Using fsutil.exe to query the Extended Attribute
KERNEL.SMARTLOCKER.ORIGINCLAIM
 - If the output shows "00" in the fifth position of the data, it indicates that the Extended Attribute is related to a **Managed Installer**.

```
PS C:\Users\JohnDoe> fsutil file queryea "C:\Program Files\7-Zip\7z.exe"

Extended Attributes (EA) information for file C:\Program Files\7-Zip\7z.exe:

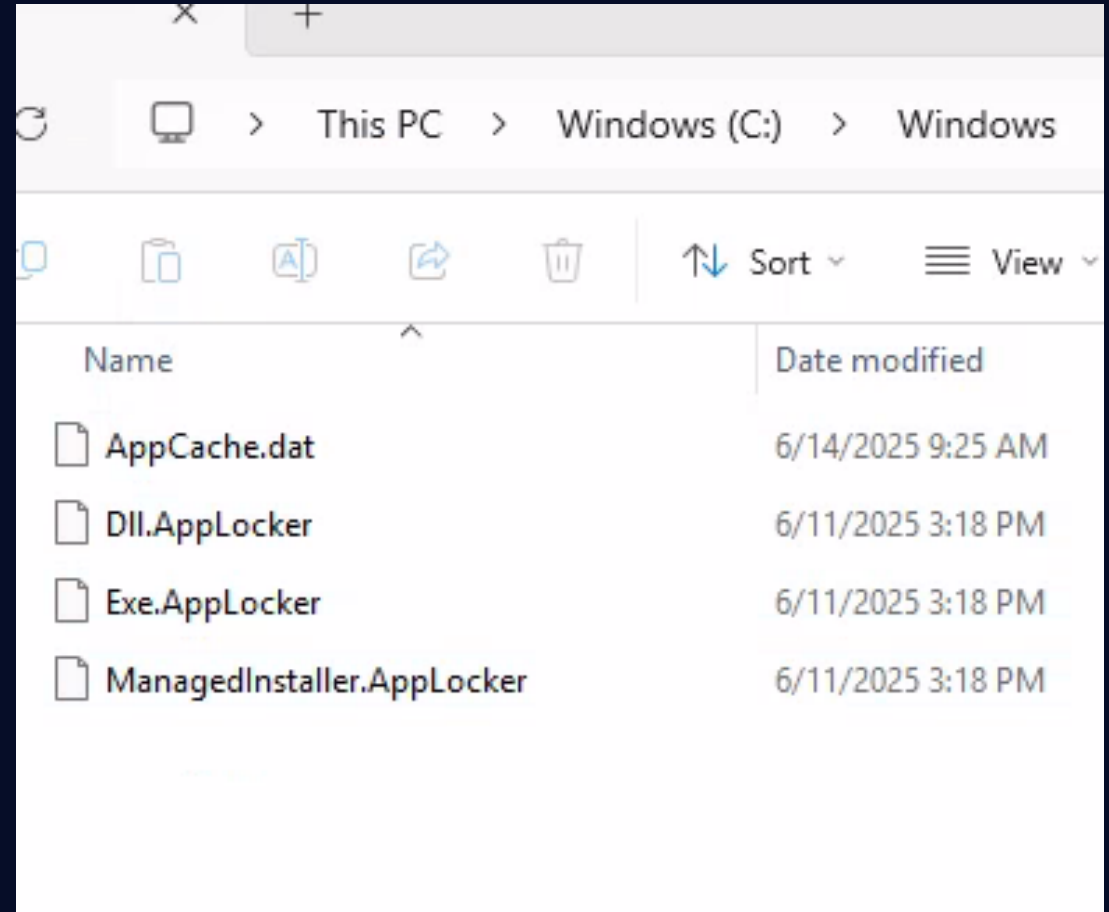
Total Ea Size: 0x224

Ea Buffer Offset: 0
Ea Name: $KERNEL.SMARTLOCKER.ORIGINCLAIM
Ea Value Length: 12c
0000:  01 00 00 00 00 00 00 00  00 00 00 00 01 00 00 00  .....
0010:  6d 15 3d 0a 22 53 c9 d0  87 01 db 48 6b 26 ab 97  m.=."S.....Hk&..
0020:  11 d0 1e 5f 17 8b 4b 0b  eb 82 24 6f fd 1d 99 7f  ..._.K...$o...
0030:  00 00 00 00 00 00 00 00  ea 00 00 00 5c 00 3f 00  .....\.?.
0040:  3f 00 5c 00 43 00 3a 00  5c 00 50 00 72 00 6f 00  ?.\.C.:.\.P.r.o.
0050:  67 00 72 00 61 00 6d 00  20 00 46 00 69 00 6c 00  g.r.a.m. .F.i.l.
0060:  65 00 73 00 20 00 28 00  78 00 38 00 36 00 29 00  e.s. (.x.8.6.).
0070:  5c 00 4d 00 69 00 63 00  72 00 6f 00 73 00 6f 00  \.M.i.c.r.o.s.o.
0080:  66 00 74 00 20 00 49 00  6e 00 74 00 75 00 6e 00  f.t. .I.n.t.u.n.
0090:  65 00 20 00 4d 00 61 00  6e 00 61 00 67 00 65 00  e. .M.a.n.a.g.e.
00a0:  6d 00 65 00 6e 00 74 00  20 00 45 00 78 00 74 00  m.e.n.t. .E.x.t.
00b0:  65 00 6e 00 73 00 69 00  6f 00 6e 00 5c 00 4d 00  e.n.s.i.o.n.\.M.
00c0:  69 00 63 00 72 00 6f 00  73 00 6f 00 66 00 74 00  i.c.r.o.s.o.f.t.
00d0:  2e 00 4d 00 61 00 6e 00  61 00 67 00 65 00 6d 00  ..M.a.n.a.g.e.m.
00e0:  65 00 6e 00 74 00 2e 00  53 00 65 00 72 00 76 00  e.n.t...S.e.r.v.
00f0:  69 00 63 00 65 00 73 00  2e 00 49 00 6e 00 74 00  i.c.e.s...I.n.t.
0100:  75 00 6e 00 65 00 57 00  69 00 6e 00 64 00 6f 00  u.n.e.W.i.n.d.o.
0110:  77 00 73 00 41 00 67 00  65 00 6e 00 74 00 2e 00  w.s.A.g.e.n.t...
0120:  65 00 78 00 65 00 00 00  00 00 00 00 00 00 00 00  e.x.e.....
```



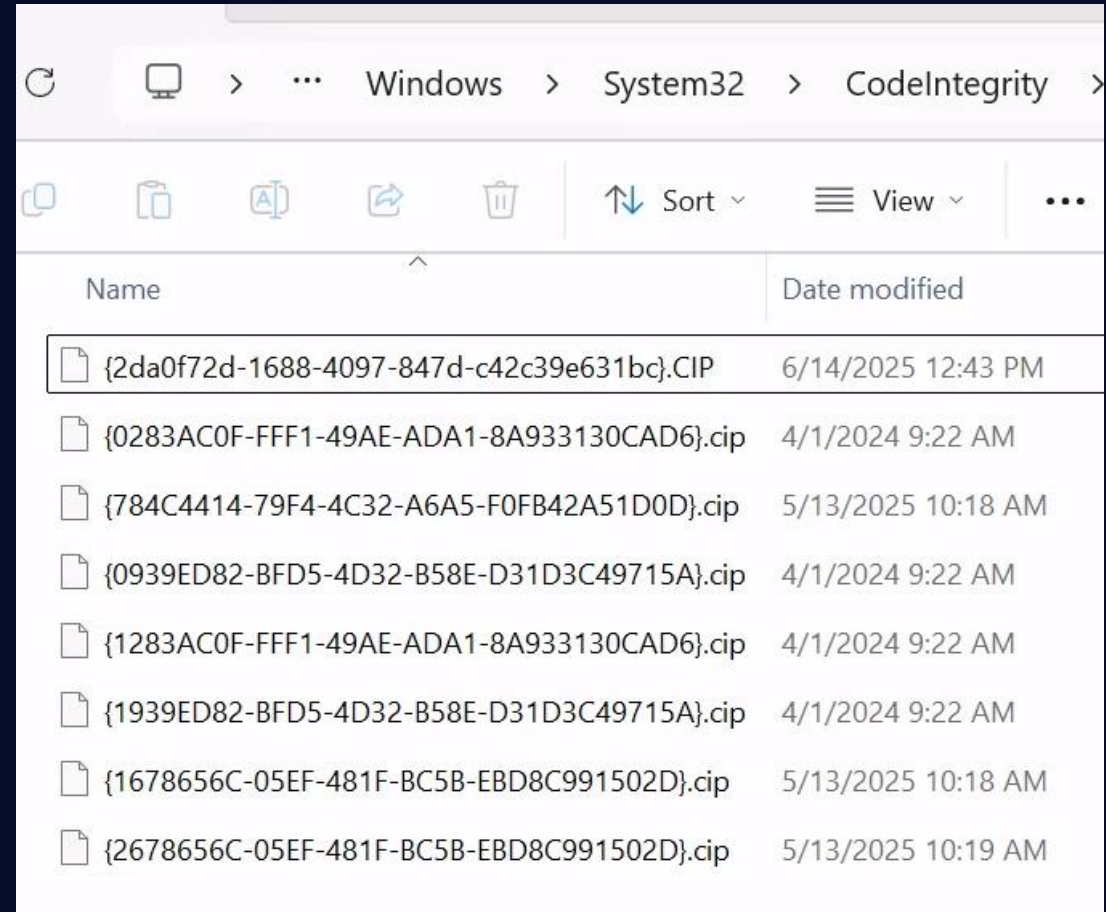
Troubleshoot using File Explorer

- Managed Installer location
 - C:\Windows\System32\AppLocker



Troubleshoot using File Explorer

- Policy location
 - C:\Windows\System32\CodeIntegrity\CiPolicies\Active



Name	Date modified
{2da0f72d-1688-4097-847d-c42c39e631bc}.CIP	6/14/2025 12:43 PM
{0283AC0F-FFF1-49AE-ADA1-8A933130CAD6}.cip	4/1/2024 9:22 AM
{784C4414-79F4-4C32-A6A5-F0FB42A51D0D}.cip	5/13/2025 10:18 AM
{0939ED82-BFD5-4D32-B58E-D31D3C49715A}.cip	4/1/2024 9:22 AM
{1283AC0F-FFF1-49AE-ADA1-8A933130CAD6}.cip	4/1/2024 9:22 AM
{1939ED82-BFD5-4D32-B58E-D31D3C49715A}.cip	4/1/2024 9:22 AM
{1678656C-05EF-481F-BC5B-EBD8C991502D}.cip	5/13/2025 10:18 AM
{2678656C-05EF-481F-BC5B-EBD8C991502D}.cip	5/13/2025 10:19 AM



Troubleshoot using the Event Viewer

- Applications and Services Logs
 - > Microsoft > Windows > DeviceManagement-Enterprise-Diagnostics-Provider > Admin
 - Event ID 814/815: Indicates MDM policy download and application status

Error	6/14/2025 8:18:33 PM	DeviceManagement-Enterprise-Diagnostics...	1904	None
Error	6/14/2025 8:18:31 PM	DeviceManagement-Enterprise-Diagnostics...	2451	None
Error	6/14/2025 8:18:31 PM	DeviceManagement-Enterprise-Diagnostics...	2451	None
Error	6/14/2025 8:18:31 PM	DeviceManagement-Enterprise-Diagnostics...	2451	None
Error	6/14/2025 8:18:31 PM	DeviceManagement-Enterprise-Diagnostics...	2451	None
Error	6/14/2025 8:18:31 PM	DeviceManagement-Enterprise-Diagnostics...	2451	None
Error	6/14/2025 8:18:31 PM	DeviceManagement-Enterprise-Diagnostics...	2451	None
Error	6/14/2025 8:18:29 PM	DeviceManagement-Enterprise-Diagnostics...	2451	None
Information	6/14/2025 8:18:22 PM	DeviceManagement-Enterprise-Diagnostics...	1920	None
Information	6/14/2025 8:18:21 PM	DeviceManagement-Enterprise-Diagnostics...	1905	None
Error	6/14/2025 8:18:13 PM	DeviceManagement-Enterprise-Diagnostics...	2451	None
Error	6/14/2025 8:18:13 PM	DeviceManagement-Enterprise-Diagnostics...	2451	None
Error	6/14/2025 8:18:13 PM	DeviceManagement-Enterprise-Diagnostics...	2451	None
Information	6/14/2025 8:18:13 PM	DeviceManagement-Enterprise-Diagnostics...	1905	None
Information	6/14/2025 8:18:12 PM	DeviceManagement-Enterprise-Diagnostics...	1912	None
Information	6/14/2025 8:18:12 PM	DeviceManagement-Enterprise-Diagnostics...	1910	None
Information	6/14/2025 8:18:12 PM	DeviceManagement-Enterprise-Diagnostics...	1904	None
Warning	6/14/2025 8:18:01 PM	DeviceManagement-Enterprise-Diagnostics...	4051	None
Error	6/14/2025 8:17:59 PM	DeviceManagement-Enterprise-Diagnostics...	1924	None
Error	6/14/2025 8:17:59 PM	DeviceManagement-Enterprise-Diagnostics...	2451	None
Error	6/14/2025 8:17:59 PM	DeviceManagement-Enterprise-Diagnostics...	2451	None
Error	6/14/2025 8:17:59 PM	DeviceManagement-Enterprise-Diagnostics...	2451	None
Error	6/14/2025 8:17:59 PM	DeviceManagement-Enterprise-Diagnostics...	2451	None
Error	6/14/2025 8:17:59 PM	DeviceManagement-Enterprise-Diagnostics...	2451	None
Error	6/14/2025 8:17:59 PM	DeviceManagement-Enterprise-Diagnostics...	2451	None
Error	6/14/2025 8:17:59 PM	DeviceManagement-Enterprise-Diagnostics...	2451	None



Troubleshoot using the Event Viewer

- Applications and Services logs
 - Microsoft - Windows - AppLocker - MSI and Script
 - Event ID 8039: Indicates that a packaged app was allowed to install or run because the policy is in audit mode. But it would have been blocked if the policy had been enforced.
 - Event ID 8040: Indicates that a packaged app was prevented from installing or running due to the policy.

MSI and Script					Number of events: 1,365 (0) New events available	
Level	Date and Time	Source	Event ID	Task Category		
Information	6/14/2025 8:12:43 PM	AppLocker	8037	None		
Information	6/14/2025 8:12:10 PM	AppLocker	8037	None		
Information	6/14/2025 8:11:35 PM	AppLocker	8037	None		
Information	6/14/2025 8:11:00 PM	AppLocker	8037	None		
Information	6/14/2025 8:10:24 PM	AppLocker	8037	None		
Information	6/14/2025 8:10:03 PM	AppLocker	8041	None		
Information	6/14/2025 8:10:03 PM	AppLocker	8037	None		
Information	6/14/2025 8:10:03 PM	AppLocker	8037	None		
Information	6/14/2025 8:09:49 PM	AppLocker	8037	None		
Information	6/14/2025 8:09:13 PM	AppLocker	8037	None		
Information	6/14/2025 8:08:40 PM	AppLocker	8037	None		
Information	6/14/2025 8:08:30 PM	AppLocker	8041	None		
Information	6/14/2025 8:08:30 PM	AppLocker	8037	None		
Information	6/14/2025 8:08:30 PM	AppLocker	8037	None		
Information	6/14/2025 8:08:30 PM	AppLocker	8041	None		
Information	6/14/2025 8:08:30 PM	AppLocker	8037	None		
Information	6/14/2025 8:08:30 PM	AppLocker	8037	None		
Information	6/14/2025 8:08:30 PM	AppLocker	8041	None		
Information	6/14/2025 8:08:30 PM	AppLocker	8037	None		
Information	6/14/2025 8:08:30 PM	AppLocker	8037	None		
Information	6/14/2025 8:08:30 PM	AppLocker	8041	None		
Information	6/14/2025 8:08:30 PM	AppLocker	8037	None		
Information	6/14/2025 8:08:30 PM	AppLocker	8037	None		
Information	6/14/2025 8:08:30 PM	AppLocker	8041	None		
Information	6/14/2025 8:08:30 PM	AppLocker	8037	None		



Troubleshoot using the Event Viewer

- Applications and Services logs
 - > Microsoft > Windows > CodeIntegrity > Operational
 - Event ID 3105: The system is attempting to refresh the Application Control policy with the specified policy ID
 - Event ID 3102: The system finished refreshing the Application Control policy.

Information	6/14/2025 8:16:47 PM	CodeIntegrity	3089 (1)
Error	6/14/2025 8:16:47 PM	CodeIntegrity	3004 (1)
Information	6/14/2025 8:16:46 PM	CodeIntegrity	3089 (1)
Error	6/14/2025 8:16:46 PM	CodeIntegrity	3004 (1)
Information	6/14/2025 8:16:45 PM	CodeIntegrity	3089 (1)
Error	6/14/2025 8:16:45 PM	CodeIntegrity	3004 (1)
Information	6/14/2025 8:16:45 PM	CodeIntegrity	3089 (1)
Error	6/14/2025 8:16:45 PM	CodeIntegrity	3004 (1)
Information	6/14/2025 8:16:45 PM	CodeIntegrity	3089 (1)
Error	6/14/2025 8:16:45 PM	CodeIntegrity	3004 (1)
Information	6/14/2025 8:16:45 PM	CodeIntegrity	3089 (1)
Error	6/14/2025 8:16:45 PM	CodeIntegrity	3004 (1)
Information	6/14/2025 8:16:45 PM	CodeIntegrity	3089 (1)
Error	6/14/2025 8:16:45 PM	CodeIntegrity	3004 (1)
Information	6/14/2025 8:16:45 PM	CodeIntegrity	3089 (1)
Error	6/14/2025 8:16:45 PM	CodeIntegrity	3004 (1)
Information	6/14/2025 8:16:45 PM	CodeIntegrity	3089 (1)
Error	6/14/2025 8:16:45 PM	CodeIntegrity	3004 (1)
Information	6/14/2025 8:16:43 PM	CodeIntegrity	3089 (1)
Error	6/14/2025 8:16:43 PM	CodeIntegrity	3004 (1)
Information	6/14/2025 8:16:43 PM	CodeIntegrity	3089 (1)
Error	6/14/2025 8:16:43 PM	CodeIntegrity	3004 (1)
Information	6/14/2025 8:16:43 PM	CodeIntegrity	3089 (1)
Error	6/14/2025 8:16:43 PM	CodeIntegrity	3004 (1)



Troubleshoot using the Event Viewer

- Applications and Services logs > Microsoft > Windows > CodeIntegrity > Operational
 - Event ID 3076: AppControlCodeIntegrityPolicyAudited: Indicates the file would have been blocked if the policy was enforced.
 - Event ID 3077: AppControlCodeIntegrityPolicyBlocked: Indicates the file didn't pass your WDAC policy and was blocked.
 - Event ID 3091: Indicates that a file didn't have managed installer authorization, and the policy is in audit mode.
 - Event ID 3099: Indicates that a policy has been loaded. This event also includes information about the options set by the Application Control policy.

Information	6/14/2025 8:16:47 PM	CodeIntegrity	3089 (1)
Error	6/14/2025 8:16:47 PM	CodeIntegrity	3004 (1)
Information	6/14/2025 8:16:46 PM	CodeIntegrity	3089 (1)
Error	6/14/2025 8:16:46 PM	CodeIntegrity	3004 (1)
Information	6/14/2025 8:16:45 PM	CodeIntegrity	3089 (1)
Error	6/14/2025 8:16:45 PM	CodeIntegrity	3004 (1)
Information	6/14/2025 8:16:45 PM	CodeIntegrity	3089 (1)
Error	6/14/2025 8:16:45 PM	CodeIntegrity	3004 (1)
Information	6/14/2025 8:16:45 PM	CodeIntegrity	3089 (1)
Error	6/14/2025 8:16:45 PM	CodeIntegrity	3004 (1)
Information	6/14/2025 8:16:45 PM	CodeIntegrity	3089 (1)
Error	6/14/2025 8:16:45 PM	CodeIntegrity	3004 (1)
Information	6/14/2025 8:16:45 PM	CodeIntegrity	3089 (1)
Error	6/14/2025 8:16:45 PM	CodeIntegrity	3004 (1)
Information	6/14/2025 8:16:45 PM	CodeIntegrity	3089 (1)
Error	6/14/2025 8:16:45 PM	CodeIntegrity	3004 (1)
Information	6/14/2025 8:16:45 PM	CodeIntegrity	3089 (1)
Error	6/14/2025 8:16:45 PM	CodeIntegrity	3004 (1)
Information	6/14/2025 8:16:45 PM	CodeIntegrity	3089 (1)
Error	6/14/2025 8:16:45 PM	CodeIntegrity	3004 (1)
Information	6/14/2025 8:16:45 PM	CodeIntegrity	3089 (1)
Error	6/14/2025 8:16:45 PM	CodeIntegrity	3004 (1)
Information	6/14/2025 8:16:43 PM	CodeIntegrity	3089 (1)
Error	6/14/2025 8:16:43 PM	CodeIntegrity	3004 (1)
Information	6/14/2025 8:16:43 PM	CodeIntegrity	3089 (1)
Error	6/14/2025 8:16:43 PM	CodeIntegrity	3004 (1)
Information	6/14/2025 8:16:43 PM	CodeIntegrity	3089 (1)
Error	6/14/2025 8:16:43 PM	CodeIntegrity	3004 (1)



Troubleshoot using the Intune Management Extension logs

- Open the HealthScripts log file.
 - C:\ProgramData\Microsoft\IntuneManagementExtension\Logs
- Look for the **Managed Installer** policy ID
 - d78c1822-e082-491a-b3a7-4a701836481e

```
<![LOG[[ManagedInstallerSync] SKU is 4.]LOG]!><time="09:30:16.5592516" date="6-14-2025" component="HealthScripts" context="" type="1" thread="23" file="">  
<![LOG[[ManagedInstaller] Found managed installer policy]LOG]!><time="09:30:16.5667755" date="6-14-2025" component="HealthScripts" context="" type="1" thread="23" file="">  
<![LOG[[ManagedInstallerSync] SKU is 4.]LOG]!><time="09:30:16.5667755" date="6-14-2025" component="HealthScripts" context="" type="1" thread="23" file="">  
<![LOG[[ManagedInstallerSync] SKU is 4.]LOG]!><time="09:30:16.5825331" date="6-14-2025" component="HealthScripts" context="" type="1" thread="23" file="">  
<![LOG[[ManagedInstaller] Set managed installer policy 1.]LOG]!><time="09:30:16.5825331" date="6-14-2025" component="HealthScripts" context="" type="1" thread="23" file="">  
<![LOG[[ManagedInstaller] Applocker policy LastWriteTime is 1]LOG]!><time="09:30:16.5825331" date="6-14-2025" component="HealthScripts" context="" type="1" thread="23" file="">  
<![LOG[[ManagedInstaller] Runner: script d78c1822-e082-491a-b3a7-4a701836481e will be executed now.]LOG]!><time="09:30:16.5825331" date="6-14-2025" component="HealthScripts" context="" type="1" thread="23" file="">  
<![LOG[[HS] ProcessScript PolicyId: d78c1822-e082-491a-b3a7-4a701836481e PolicyType: 10]LOG]!><time="09:30:16.5984325" date="6-14-2025" component="HealthScripts" context="" type="1" thread="23" file="">  
<![LOG[[ManagedInstallerSync] SKU is 4.]LOG]!><time="09:30:16.6079905" date="6-14-2025" component="HealthScripts" context="" type="1" thread="23" file="">  
<![LOG[[ManagedInstallerSync] SKU is 4.]LOG]!><time="09:30:16.6143793" date="6-14-2025" component="HealthScripts" context="" type="1" thread="23" file="">
```



Troubleshoot using KQL (Device Query)

WindowsEvent("Microsoft-Windows-CodeIntegrity/Operational",1h)
| where EventId == 3004

LogName ⓘ	EventId ⓘ	Level ⓘ	LoggedDateTime ⓘ	Message ⓘ	ProviderName ⓘ	WindowsUserAccount ⓘ
Microsoft-Window...	3089	INFORMATION	2025-06-14T18:04:...	Signature informati...	Microsoft-Window...	NT AUTHORITY\SYSTEM
Microsoft-Window...	3004	ERROR	2025-06-14T18:04:...	Windows is unable...	Microsoft-Window...	NT AUTHORITY\SYSTEM
Microsoft-Window...	3089	INFORMATION	2025-06-14T18:04:...	Signature informati...	Microsoft-Window...	NT AUTHORITY\SYSTEM
Microsoft-Window...	3004	ERROR	2025-06-14T18:04:...	Windows is unable...	Microsoft-Window...	NT AUTHORITY\SYSTEM
Microsoft-Window...	3089	INFORMATION	2025-06-14T18:04:...	Signature informati...	Microsoft-Window...	NT AUTHORITY\SYSTEM



Best Practices for App Control for Business

Deploy Managed Installer first so that apps can start getting the Managed Installer tag.

That way, if later you are ready to deploy App Control policy, you can use “Trust apps from managed installer” rule without having to redeploy apps.

1

When the admin completes opt-in for a tenant, ensure caching time has passed.

Any change to add and remove the AppLocker policy could take up to an hour to be effective on the device.

2

In terms of App Control policy, use audit mode first and then switch to enforce mode.

- Audit mode will allow you to monitor the app behavior and block events without affecting the end user experience.
- Enforce mode will enforce the app allow list and block any apps that are not allowed.

3

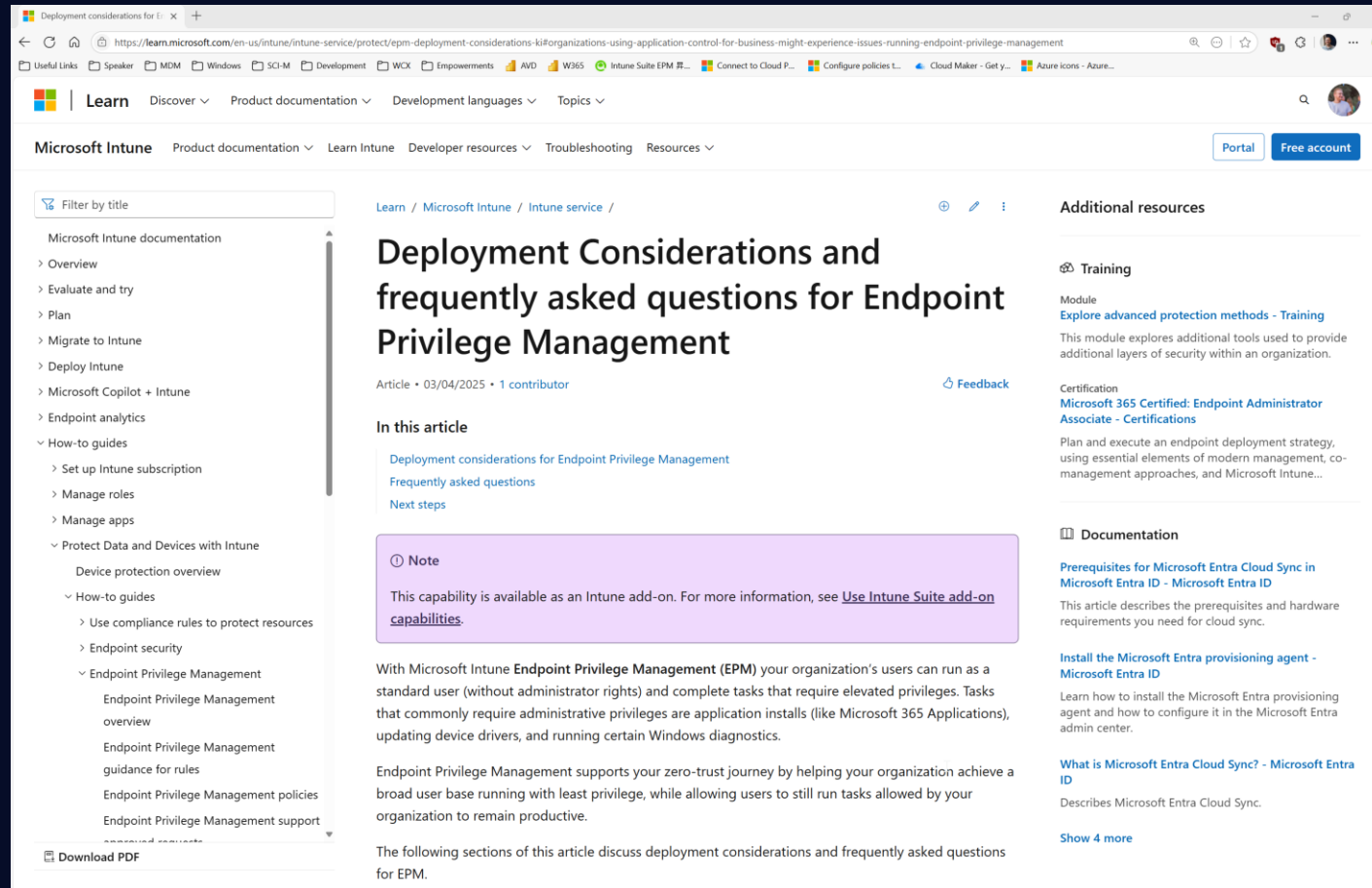
Intune just helps facilitate the App Control policy deployment and reporting. The actual app behavior and block events are handled by Windows and Defender.

Therefore, it is important to understand the App Control for Business feature and how it works with AppLocker and Managed Installer.



Deployment considerations for Endpoint Privilege Management

What can go wrong...



The screenshot shows a web browser displaying the Microsoft Intune documentation page. The page title is "Deployment Considerations and frequently asked questions for Endpoint Privilege Management". The breadcrumb trail is "Learn / Microsoft Intune / Intune service /". The page is dated "03/04/2025" and has "1 contributor".

In this article

- [Deployment considerations for Endpoint Privilege Management](#)
- [Frequently asked questions](#)
- [Next steps](#)

Note

This capability is available as an Intune add-on. For more information, see [Use Intune Suite add-on capabilities](#).

With Microsoft Intune **Endpoint Privilege Management (EPM)** your organization's users can run as a standard user (without administrator rights) and complete tasks that require elevated privileges. Tasks that commonly require administrative privileges are application installs (like Microsoft 365 Applications), updating device drivers, and running certain Windows diagnostics.

Endpoint Privilege Management supports your zero-trust journey by helping your organization achieve a broad user base running with least privilege, while allowing users to still run tasks allowed by your organization to remain productive.

The following sections of this article discuss deployment considerations and frequently asked questions for EPM.

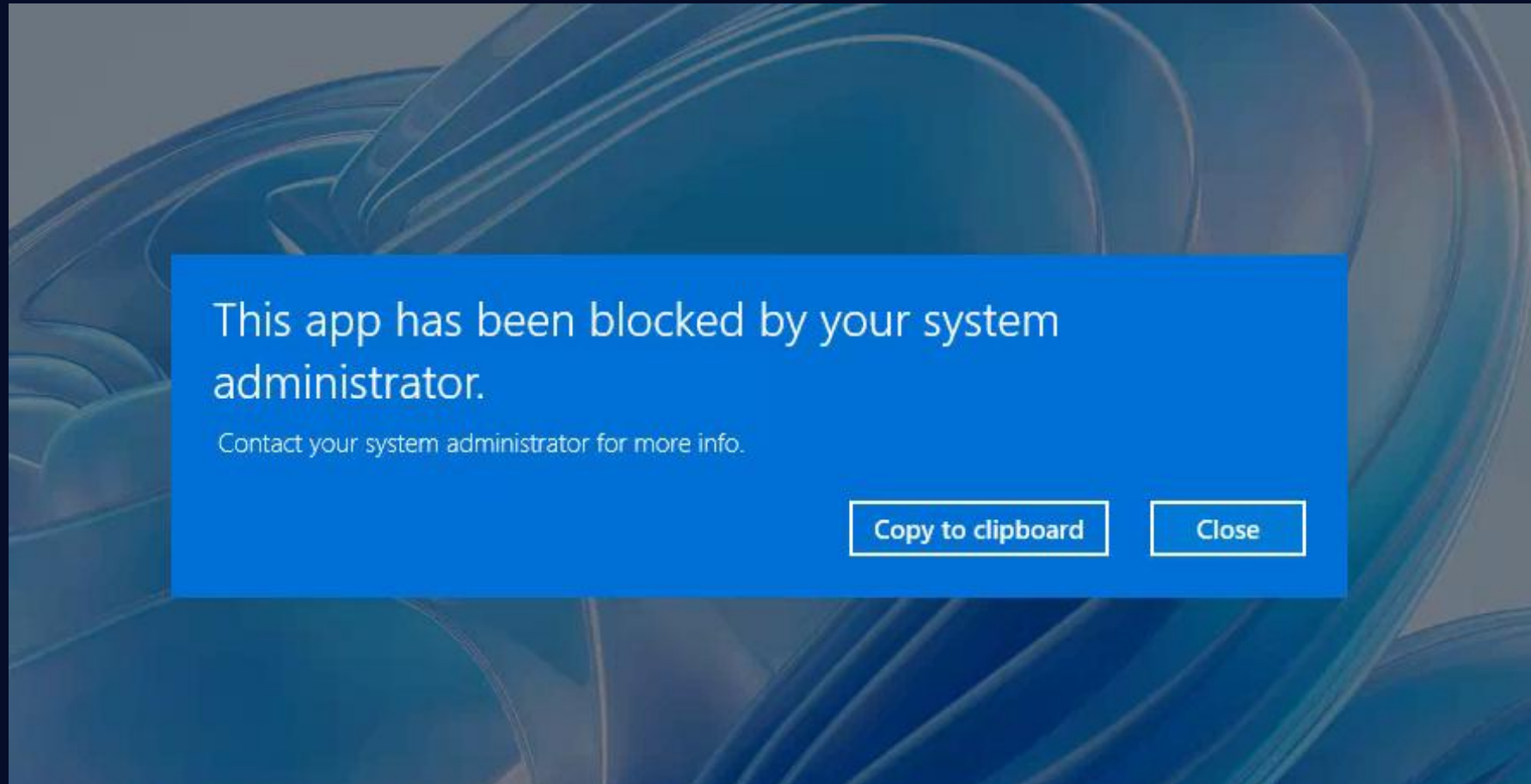
Additional resources

- Training**
 - Module: [Explore advanced protection methods - Training](#)
This module explores additional tools used to provide additional layers of security within an organization.
- Certification**
 - [Microsoft 365 Certified: Endpoint Administrator Associate - Certifications](#)
Plan and execute an endpoint deployment strategy, using essential elements of modern management, co-management approaches, and Microsoft Intune...
- Documentation**
 - [Prerequisites for Microsoft Entra Cloud Sync in Microsoft Entra ID - Microsoft Entra ID](#)
This article describes the prerequisites and hardware requirements you need for cloud sync.
 - [Install the Microsoft Entra provisioning agent - Microsoft Entra ID](#)
Learn how to install the Microsoft Entra provisioning agent and how to configure it in the Microsoft Entra admin center.
 - [What is Microsoft Entra Cloud Sync? - Microsoft Entra ID](#)
Describes Microsoft Entra Cloud Sync.

[Show 4 more](#)



Monitoring the impact of EPM on App Control for Business



Monitoring the impact of EPM on App Control for Business

Event 3004, CodeIntegrity

General Details

Windows is unable to verify the image integrity of the file \Device\HarddiskVolume4\Program Files\Microsoft EPM Agent\EPMAAdapter\EpmAdapter.dll because file hash could not be found on the system. A recent hardware or software change might have installed a file that is signed incorrectly or damaged, or that might be malicious software from an unknown source.

Log Name: Microsoft-Windows-CodeIntegrity/Operational
Source: CodeIntegrity Logged: 6/14/2025 8:35:01 PM
Event ID: 3004 Task Category: (1)
Level: Error Keywords:
User: SYSTEM Computer: EU-DEMO-8AF5Z
OpCode: (6815744)



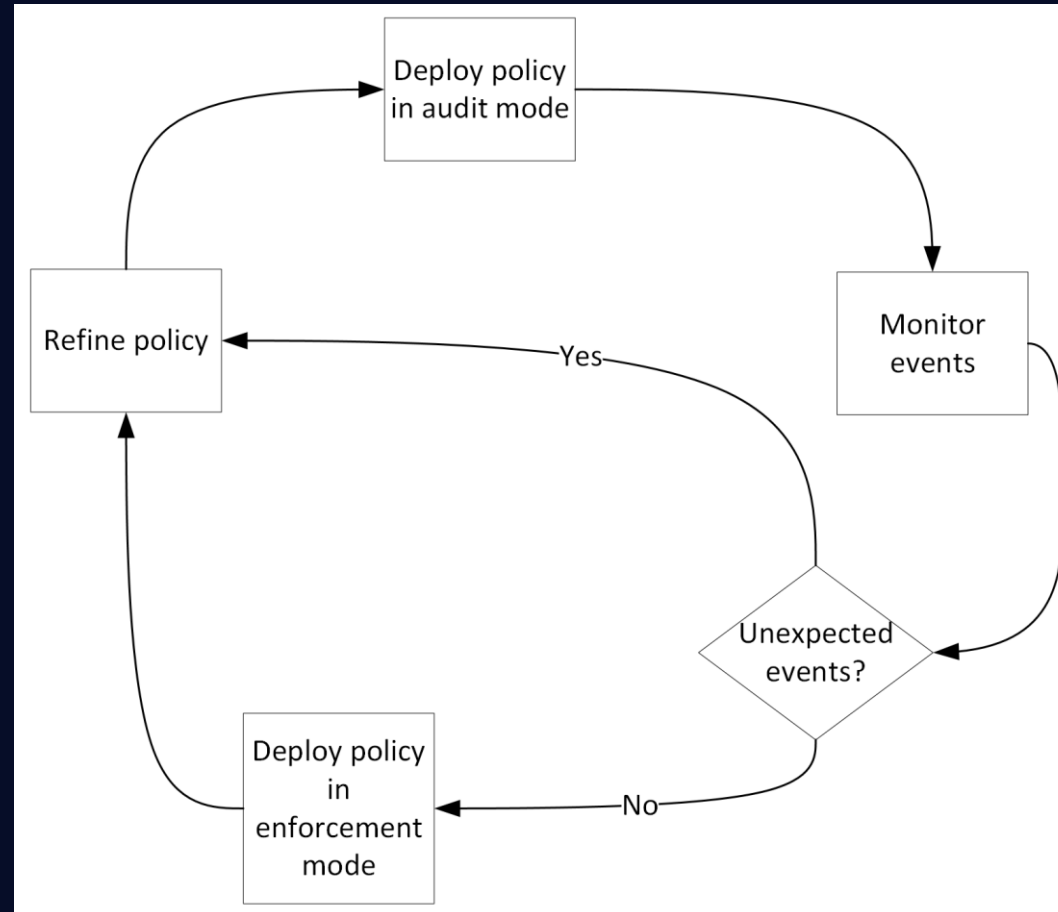
Remediating the impact of EPM

- Creating a Supplemental Policy on top of your Base Policy!

```
<Rules>
  <Rule>
    <Option>Enabled:Unsigned System Integrity Policy</Option>
  </Rule>
</Rules>
<EKUs/>
<FileRules>
  <Allow ID="ID_ALLOW_A_0197729C4E2370169908EAA178294674" FriendlyName="Sha256 Hash for: EPMAdapter.mof" Hash="B866CCFFC8D1ADF358CF6377EC8DE7C35CFB683F11E53B288EDA69391F59F1CC"/>
  <Allow ID="ID_ALLOW_A_0197729C4E23735388F42948C350F93E" FriendlyName="Sha1 Hash for: EPMAdapter.mof" Hash="BC5F5DE76A5072C881C31EB153F793C6DE52C59F"/>
  <Allow ID="ID_ALLOW_A_0197729C4E237A2E93396404D652860" FriendlyName="Sha256 Hash for: EPMGlobalSettingsAdapter.mof" Hash="CF9910DAF97F8C34D6310A787F05830A2C5CDB6246339F06A8D7E004707BA9EB"/>
  <Allow ID="ID_ALLOW_A_0197729C4E2378488B2F727F18F80810" FriendlyName="Sha1 Hash for: EPMGlobalSettingsAdapter.mof" Hash="16904D0D9CFB1C3D6D6A4750EC7BF7E0728199E"/>
  <Allow ID="ID_ALLOW_A_0197729C4E237A41A158CD46F6D81397" FriendlyName="Sha256 Hash for: EPMGlobalSettingsAdapter.mof" Hash="811C3B7B93478376333C4EAC693EEDAA69B1BE176EACF82F6BD0F8EE0996364B"/>
  <Allow ID="ID_ALLOW_A_0197729C4E2375ADA408B85B3AC6750" FriendlyName="Sha1 Hash for: EPMGlobalSettingsAdapter.mof" Hash="FFB34CA62E9BD87A0746A7E65E8F41A7B6BA5CA"/>
  <Allow ID="ID_ALLOW_A_0197729C4E237E628C080A1CF6B1F145" FriendlyName="Sha256 Hash for: EPMClientSettingsAdapter.mof" Hash="830421261C739E9C7D3BE324B8420FA2421A7A6E297145364548306F0C6034E8"/>
  <Allow ID="ID_ALLOW_A_0197729C4E2371A58177101279F788D" FriendlyName="Sha1 Hash for: EPMClientSettingsAdapter.mof" Hash="780750CD2B0274E239E355822C6811CAB863971A"/>
  <Allow ID="ID_ALLOW_A_0197729C4E23703F9CA72A0E655208872" FriendlyName="Sha256 Hash for: EPMClientSettingsAdapter.mof" Hash="F5655C135DDAC09AC78BB6A4546A20AF27F60A918984BAF1C8BB6FC36D081375"/>
  <Allow ID="ID_ALLOW_A_0197729C4E2371569A5AFDABEACB1EA3" FriendlyName="Sha1 Hash for: EPMClientSettingsAdapter.mof" Hash="ECC09F1BD5B2822689264F706E80437A0DD09C88"/>
  <Allow ID="ID_ALLOW_A_0197729C4E23734EAE9F68F06D72248" FriendlyName="Sha256 Hash for: EPMAdapterNs.mof" Hash="9F72C9144C35008FDA6206C724502C56620223F278EA03C1F033311027098CC3"/>
  <Allow ID="ID_ALLOW_A_0197729C4E237C1C84A305FBA440F784" FriendlyName="Sha1 Hash for: EPMAdapterNs.mof" Hash="1A7B24443D5C378305B24C81D44323085BA83659"/>
</FileRules>
<Signers>
  <Signer ID="ID_SIGNER_A_0197729C4E23738AA1C208E5C2083CFC" Name="Microsoft Code Signing PCA 2011">
    <CertRoot Type="TBS" Value="F6F717A43AD9ABDDC8CEFD0E1C505462535E7D1307E630F9544A2D14FE8BF26E"/>
    <CertPublisher Value="Microsoft Corporation"/>
  </Signer>
  <Signer ID="ID_SIGNER_A_0197729C4E237DFB07A2F63326D2405" Name="Microsoft Code Signing PCA 2010">
    <CertRoot Type="TBS" Value="121AF48922A74247E449DF500E37609CC1451A1FE06B2C87E1E079B492B08195"/>
    <CertPublisher Value="Microsoft Corporation"/>
  </Signer>
  <Signer ID="ID_SIGNER_A_0197729C4E237EECA2EC0CA68FDCFA19" Name=".NET Foundation Projects Code Signing CA">
    <CertRoot Type="TBS" Value="B6D27A37A95C0A174C66496B3225B11A62508468BC31FCA10D69D61F92DF5084"/>
    <CertPublisher Value="Reactive Extensions for .NET (.NET Foundation)"/>
  </Signer>
  <Signer ID="ID_SIGNER_A_0197729C4E23767DAE5FFB88F539F008" Name="Microsoft Code Signing PCA 2011">
    <CertRoot Type="TBS" Value="F6F717A43AD9ABDDC8CEFD0E1C505462535E7D1307E630F9544A2D14FE8BF26E"/>
    <CertPublisher Value="Microsoft 3rd Party Application Component"/>
  </Signer>
  <Signer ID="ID_SIGNER_A_0197729C4E23736781748E4D16592F12" Name="Microsoft Code Signing PCA">
    <CertRoot Type="TBS" Value="27543A3F7612D2E61C7228321722402F63A07DE"/>
    <CertPublisher Value="Microsoft Corporation"/>
  </Signer>
  <Signer ID="ID_SIGNER_A_0197729C4E2377018EA4437C6FE91C58" Name="Microsoft Code Signing PCA 2011">
    <CertRoot Type="TBS" Value="F6F717A43AD9ABDDC8CEFD0E1C505462535E7D1307E630F9544A2D14FE8BF26E"/>
    <CertPublisher Value=".NET"/>
  </Signer>
  <Signer ID="ID_SIGNER_A_0197729C4E2370458CAD08672CE1B284" Name="Microsoft Code Signing PCA 2011">
    <CertRoot Type="TBS" Value="F6F717A43AD9ABDDC8CEFD0E1C505462535E7D1307E630F9544A2D14FE8BF26E"/>
    <CertPublisher Value="Microsoft Corporation"/>
  </Signer>
  <Signer ID="ID_SIGNER_A_0197729C4E2374878497530E14A83F09" Name="Microsoft Windows Third Party Component CA 2012">
    <CertRoot Type="TBS" Value="CEC1AFD0E310C55C1DCC601A88E172917706AA32FB5EAF826813547FDF02DD46"/>
    <CertPublisher Value="Microsoft Windows Hardware Compatibility Publisher"/>
  </Signer>
</Signers>
```

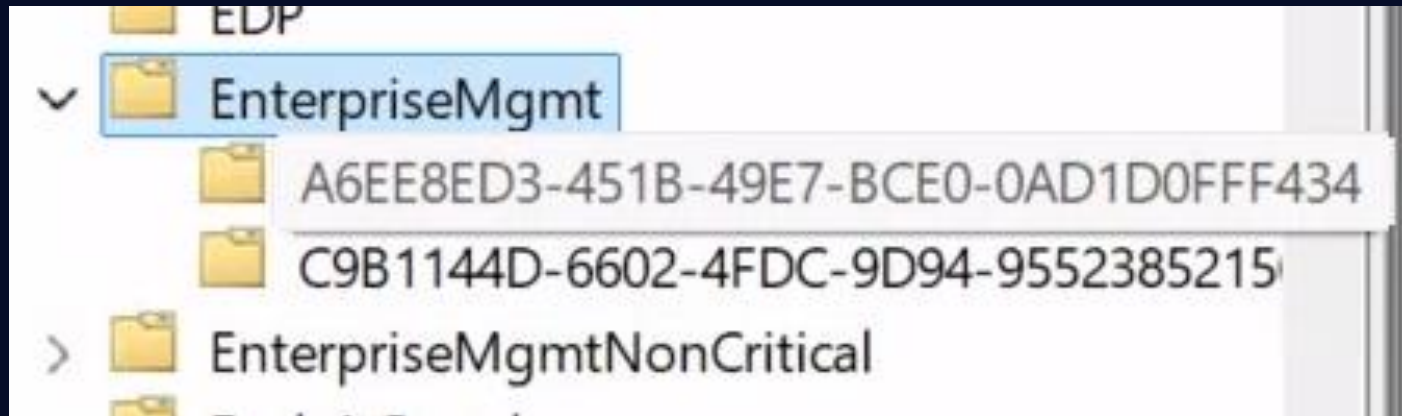


Creating the XML

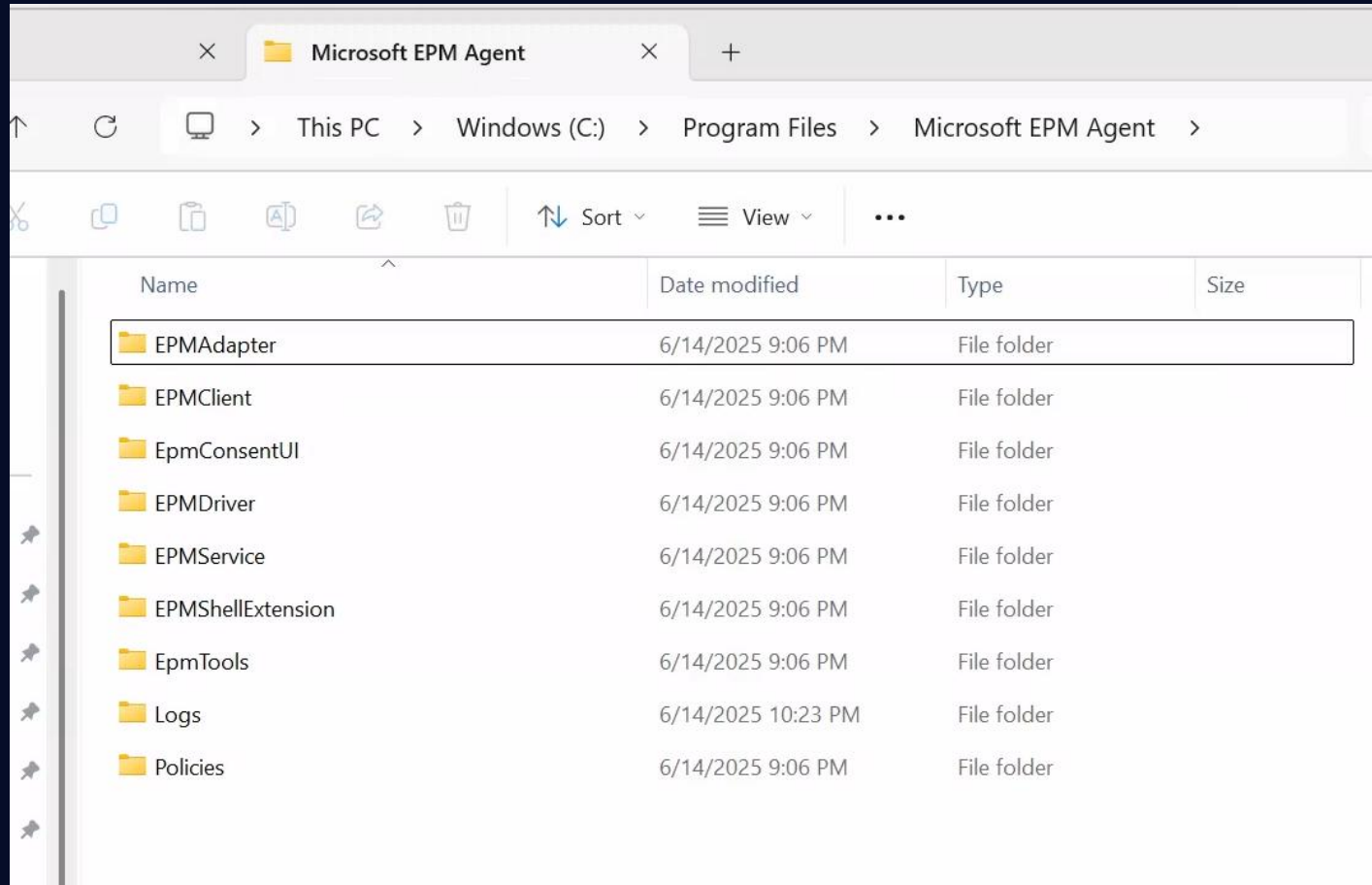


Troubleshooting EPM – Confirm Dual Enrollment

- Task Scheduler > Microsoft > Windows > EnterpriseMgmt > EnterpriseMgmt
 - 2 GUIDs should be visible



Troubleshooting EPM – Check the files



The screenshot shows a Windows File Explorer window with the address bar set to 'This PC > Windows (C:) > Program Files > Microsoft EPM Agent'. The file list is displayed in a table with columns for Name, Date modified, Type, and Size. The 'EPMAAdapter' folder is selected. The table contains the following data:

Name	Date modified	Type	Size
EPMAAdapter	6/14/2025 9:06 PM	File folder	
EPMClient	6/14/2025 9:06 PM	File folder	
EpmConsentUI	6/14/2025 9:06 PM	File folder	
EPMDriver	6/14/2025 9:06 PM	File folder	
EPMService	6/14/2025 9:06 PM	File folder	
EPMShellExtension	6/14/2025 9:06 PM	File folder	
EpmTools	6/14/2025 9:06 PM	File folder	
Logs	6/14/2025 10:23 PM	File folder	
Policies	6/14/2025 9:06 PM	File folder	



Troubleshooting EPM – Verify the EPM Service

- Open Task Manager
- Verify EPM Service

 dwm.exe	1280	Running	DWM-1	00	7,936 K	x64	Desktop Window Manager
 dwm.exe	6432	Running	DWM-2	00	48,620 K	x64	Desktop Window Manager
 EpmService.exe	4012	Running	SYSTEM	00	19,172 K	x64	EpmService
 explorer.exe	8556	Running	JohnDoe	00	120,832 K	x64	Windows Explorer
 fontdrvhost.exe	1068	Running	UMFD-1	00	1,200 K	x64	Usermode Font Driver Host



Troubleshooting EPM - PowerShell



```
Import-Module 'C:\Program Files\Microsoft EPM Agent\EpmTools\EpmCmdlets.dll'  
Get-Policies -PolicyType ElevationRules -Verbose | fl  
Get-Policies -PolicyType ClientSettings -Verbose | fl
```



Troubleshooting EPM - PowerShell

```
VERBOSE: Retrieving EPM Agent policies information for policy type 'ElevationRules'.  
VERBOSE: Policies processed by the ElevationRules adapter, count=3
```

```
ExeFileNames      : NP++.exe  
PolicyId           : a65be32d-5f36-4e58-8768-267560a659f0  
Version            : 38907E6A01013FF53E1F248414894BE11B8D711828132B3C51C8B1EDD0914C91  
LastWrite          : 6/14/2025 9:06:37 PM  
LastVerifiedTimestamp : 6/14/2025 10:14:18 PM  
UserSids           : S-1-12-1-3922581860-1328325212-3485747366-2170649095
```

```
ExeFileNames      : wireshark.exe  
PolicyId           : 5eafadce-800b-48fc-a56f-b73b10e09908  
Version            : 4E034F1846AD00A4B207AADDA1BA28E7FFF8B254898B2A9364B5B9A8B5F59E42  
LastWrite          : 6/14/2025 9:06:16 PM  
LastVerifiedTimestamp : 6/14/2025 10:14:02 PM  
UserSids           : S-1-12-1-3922581860-1328325212-3485747366-2170649095
```

```
ExeFileNames      :  
PolicyId           : d6f86036-91a7-4253-a893-9dcd9ecf2686  
Version            : D072B33FA651A37CAE85215AC571D78D16F62893D6CC75F28AE413B951C44EDE  
LastWrite          : 6/14/2025 9:06:16 PM  
LastVerifiedTimestamp : 6/14/2025 10:14:02 PM  
UserSids           : S-1-12-1-3922581860-1328325212-3485747366-2170649095
```

```
VERBOSE: Retrieving EPM Agent policies information for policy type 'ClientSettings'.  
VERBOSE: Policies processed by the ClientSettings adapter, count=3
```

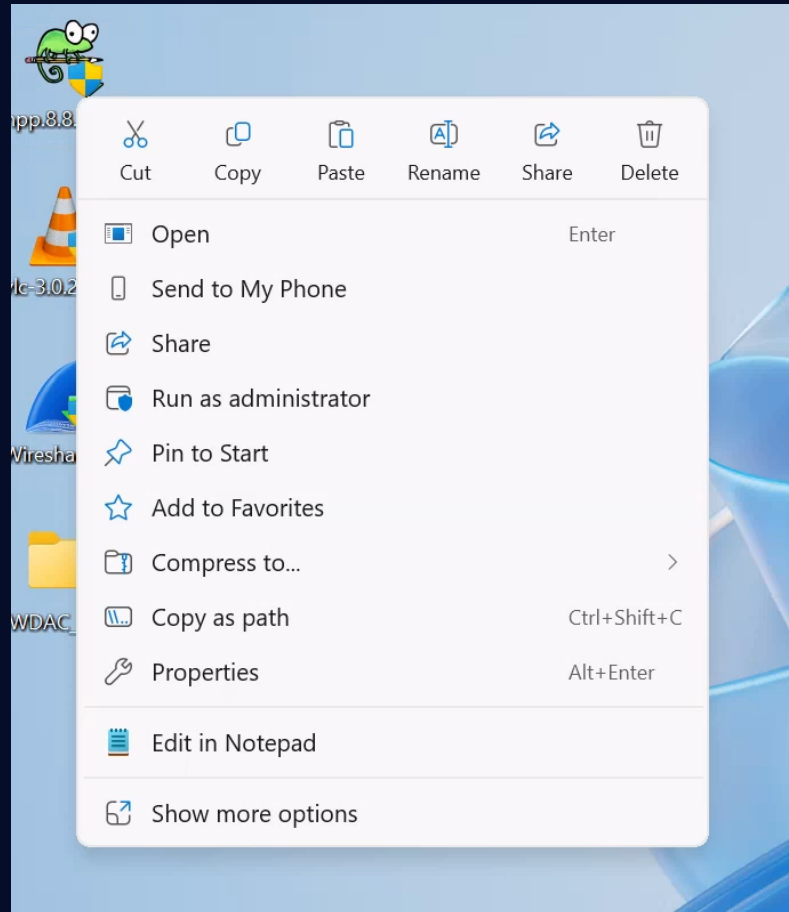
```
Settings           : {[EnableEpm, True]}  
PolicyId           : 9975a6f1-9716-475c-9ff0-a46e58badcc1  
Version            : CBEB4F1803612544A984D3C019BA0644AAF4DBDC0B1E8E9963D15A38CFB5C9F1  
LastWrite          : 6/14/2025 9:55:13 PM  
LastVerifiedTimestamp : 6/14/2025 10:14:18 PM  
UserSids           : S-1-5-18
```

```
Settings           : {[TelemetryLevel, Full]}  
PolicyId           : 82dc39b0-b140-4e87-9168-892f3f237f1a  
Version            : 03E2B49B64BF559DF8F911399F0EE79E89537F3A2C5090AF6598809298A7C9EB  
LastWrite          : 6/14/2025 9:55:13 PM  
LastVerifiedTimestamp : 6/14/2025 10:14:18 PM  
UserSids           : S-1-5-18
```

```
Settings           : {[DefaultBehavior, SupportArbitrated], [ElevationValidation, None]}  
PolicyId           : 3ad13097-4525-4ad1-b050-e797cd89083e  
Version            : 18C5B69B9EF480D937B26EA594B900EFF78D5BA1FC97B14BB CD20FC433D6261A  
LastWrite          : 6/14/2025 9:55:13 PM  
LastVerifiedTimestamp : 6/14/2025 10:14:18 PM  
UserSids           : S-1-5-18
```



We're not in Kansas anymore...



Possible future solutions?

Disable App
Control?

Disable EPM?

Have the option to
add EPM as a
Managed Installer
in Intune?





